Eggshell–Agar Biocomposites

Development and characterisation for sustainable interior applications

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Background

Point of departure

Eggshell arises continuously as a food-processing by-product and is predominantly landfilled. It accounts for roughly a tenth of the mass of every egg, so the stream is neither small nor intermittent. Chemically it is almost pure calcium carbonate held in a porous, protein-bound crystalline structure, with a specific gravity below that of most mineral fillers used in construction.

Its composition and its uninterrupted supply have made it an established secondary mineral resource, and a substantial body of work has incorporated it into construction and polymer materials.

William & Owen (1995) · Torres et al. (2010) · Owuamanam & Cree (2020)

Material basis

94–97%

calcium carbonate by mass

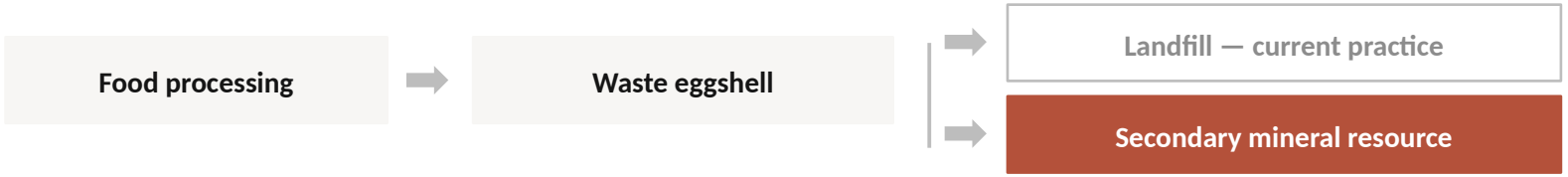
0.85–2.66

specific gravity, below most mineral fillers

> 12

months of single-source collection for this study

Disposal and valorisation routes



The case for eggshell as a mineral phase is settled. This paper is concerned with the phase that binds it.

Prior applications

Established uses of eggshell waste in engineered materials

Application	Binding phase	Reference
Partial cement replacement in concrete	Portland clinker	Teara et al. (2019); Bhartiya & Dubey (2018)
Compressed laterite brick	Soil – lime	Adogla et al. (2016)
Hot-mix asphalt	Bitumen	Razzaq & Yousif (2017)
Clay stabilisation	Clay mineral	Soundara & Vilasini (2015)
Eco-ceramic wall tile	Ceramic body	Vilarinho et al. (2022)
Particulate filler in PVC	Poly(vinyl chloride)	Skórczewska et al. (2020)
Filler in flexible polyurethane foam	Polyurethane	Victor et al. (2019)
Ceramic–metal composite	Metal matrix	Bilici et al. (2023); Yönetken et al. (2016)
Bone tissue scaffold	Calcium phosphate	Shafiei et al. (2019)
Wallpaper	Cellulosic furnish	Tutuş et al. (2018)

What every row above has in common

MINERAL PHASE · **recovered from waste in every study above**

BINDING PHASE · **cementitious, ceramic or petrochemical in every study above**

Agar as binder

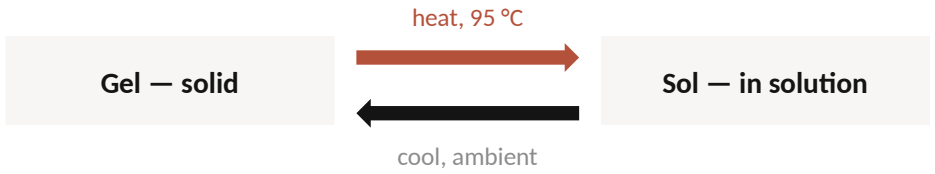
Properties and existing characterisation

Agar is a thermoreversible polysaccharide extracted from red seaweed. It melts on heating and sets on cooling without chemical crosslinking, so no reactive hardener is required. Its behaviour with glycerol has been mapped systematically across 32 formulations and released as an open reference dataset.

Scope of that characterisation

Unfilled thin films, loaded in uniaxial tension, evaluated for packaging. Agar has not been characterised as the binding phase of a mineral-filled cast solid in bending.

Thermoreversible setting



The transition is physical, not chemical: no reactive hardener, no cure temperature, and at 66% mineral loading no coupling agent.

Form and loading case in this study

—	Film literature	This study
Form	Thin cast film	Cast solid, 20–50 mm
Mineral phase	Unfilled	66% of solids
Loading	Uniaxial tension	Compression and bending
Intended use	Packaging	Interior surfaces

The compositional logic transfers. The form and the loading case do not, and it is the loading case that governs use as an interior surface.

Hernández et al. (2022), Materials 15(11), 3954 — 32 agar / glycerin / water formulations, released as an open dataset.

Research gap

Three gaps follow from the preceding two slides

01

The binding phase is never biological

Across the eggshell literature the mineral is recovered from waste while the matrix remains Portland clinker, bitumen, lime, a ceramic body or a thermoplastic. No study reports an eggshell composite in which both phases are of biological origin and fully biodegradable.

02

Agar is characterised only as an unfilled film

The reference dataset covers thin films in uniaxial tension for packaging. The behaviour of agar as the binding phase of a highly mineral-filled cast solid, loaded in bending, is not reported.

03

Property sets are reported in fragments

Compressive, flexural, reaction-to-fire and hygric behaviour are reported in separate studies on separate formulations. No single study gives all four for one eggshell–biopolymer formulation family.

The contribution is therefore a documented protocol and a single coherent property set, not a novel constituent.

Aim

To document a reproducible production protocol for a fully bio-based eggshell–agar composite, and to characterise its mechanical, reaction-to-fire and hygric behaviour against published test standards.

Objectives

01	**Document the protocol in full**	Every parameter of mineral preparation, binder preparation, casting and cure reported in a form another laboratory can repeat.
02	**Characterise against published standards**	Compressive and flexural behaviour, ignitability and water absorption, determined to ASTM D790-17, ISO 11925-2:2020 and ASTM D570-98.
03	**Position the results**	Benchmarked against conventional interior materials and against the bio-based composite family, from published property data.
04	**Derive an evaluation criterion**	Establish which measured quantity governs service behaviour for this material class, and report it alongside strength.

Scope

Within scope

Dimension	This study
Formulations	Five — three compressive, two flexural
Specimens	Three per formulation for mechanical testing
Properties	Compression, flexure, ignitability, water absorption, bulk density
Standards	ASTM D790-17 · ISO 11925-2:2020 · ASTM D570-98(2018)
Conditions	Ambient cure, single laboratory, single operator
Application	Non-loadbearing interior surfaces

Outside scope

No durability, classification or life-cycle data were generated. The results reported here are characterisation data, not a performance specification.

Each exclusion is revisited as a defined next experiment in the closing section.

Property coverage

Compression | **Flexure** | **Ignitability** | **Water absorption** | **Bulk density** | Durability | Fire classification | Life-cycle | Structural use

Filled: determined in this study. Outlined: not determined, and therefore outside every claim made in this presentation.

Method

A material-driven sequence in four stages

1 → 2 → 3 → 4

Produce

Develop the composite and record every production parameter.

Iterative formulation over twelve months

Measure

Characterise the cured material against published test standards.

Compression · flexure · ignitability · absorption

Position

Benchmark the results against two material families.

Conventional interior materials · bio-based composites

Propose

Derive an evaluation criterion from the observed behaviour.

Deformation capacity reported alongside strength

The order is material-driven: properties are measured first and the application envelope is derived from them, rather than the material being selected to satisfy a predetermined form. The design proposal presented later follows from the measurements and not the reverse.

Material-driven design after Karana et al. (2015)

Research questions

01	**Which formulation variable governs compressive strength?**	Formulation
02	**How does plasticiser content affect flexural behaviour?**	Formulation
03	**How does the composite compare with conventional interior materials?**	Benchmark
04	**How does it compare with other bio-based composites?**	Benchmark
05	**Is compressive strength an adequate evaluation criterion for this class?**	Criterion

Questions one to four are answered by measurement. Question five is answered by argument from those measurements.

Hypothesis

A fully bio-based eggshell–agar composite can be produced repeatably under ambient conditions, and its performance as an interior surface material is governed by deformation capacity rather than by compressive strength.

Two clauses, tested separately

Clause 1	**Producible repeatably under ambient conditions**	Tested by the production protocol and the formulation series: a defined procedure, a fixed shell-to-binder ratio, five formulations and a cure requiring no thermal input after mixing.	Slides 12–15
Clause 2	**Governed by deformation capacity rather than strength**	Tested by the mechanical results and the two benchmarks: the failure mode in compression, the flexural response, and the position of the modulus of resilience relative to strength.	Slides 17–24

Provenance. The dissertation from which this work derives states a broader hypothesis framed on spatial resilience in interior design. The formulation above is the materials-engineering component of it, restated within the scope of the tests reported here.



Four constituents, each biodegradable

Waste eggshell powder · agar · glycerol · water, with a shellac finish. Colour and texture vary with natural pigment and mould geometry at constant composition.

Stage 1 · Mineral phase

Progressive size reduction of the shell to a sieved powder



Whole shell	Cleaned	Heat-treated	Milled	Sieved
collected from a single bakery	membrane and residue removed manually	decontamination and drying	laboratory mechanical grinder	passing fraction retained
—	—	**120 °C, 20 min**	**to homogeneous powder**	**150 µm mesh**

Particle size rationale

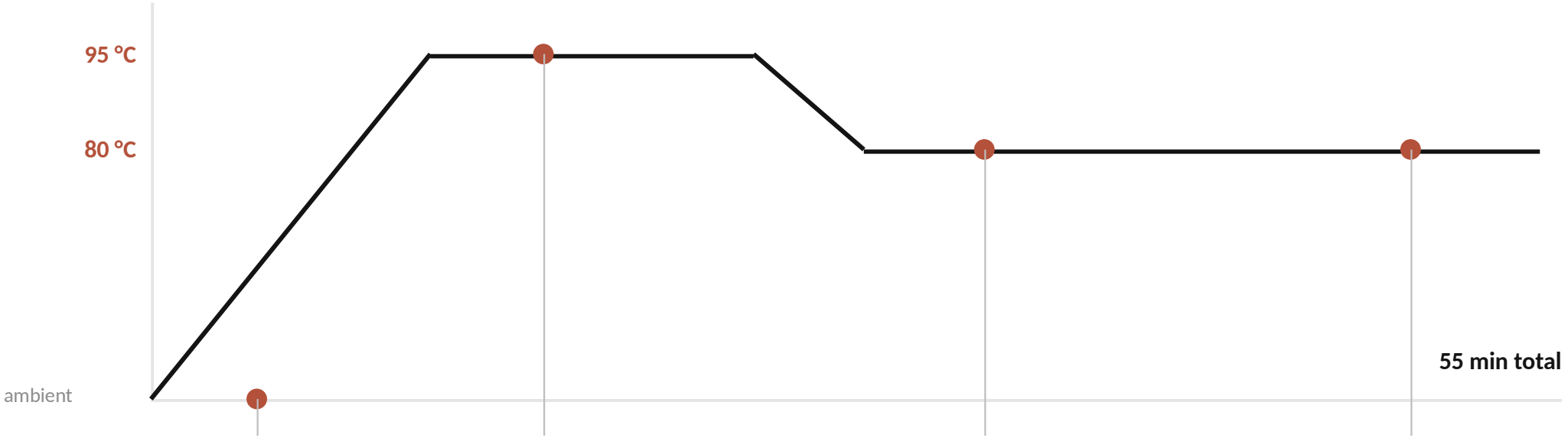
The 150 µm fraction was selected to match the particle size used in the eggshell polymer-composite literature, so that the mineral phase in this study is comparable with that body of work.

Terminology

The 120 °C step is reported as decontamination and drying, not sterilisation. Dry-heat sterilisation requires substantially higher temperatures for longer, and sterility was not verified.

Stage 2 · Binder preparation

Temperature–time profile of the 55-minute preparation

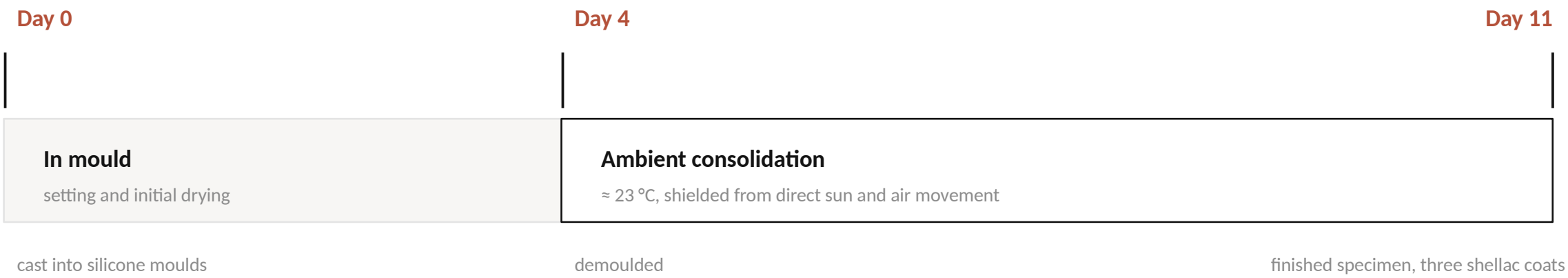


Agar is taken into solution before viscosity develops; incremental addition prevents agglomeration, and the shell fraction is introduced late to prevent sedimentation.

Schematic. The two set points and the total heating time are recorded; intermediate ramp rates are not, and are drawn indicatively. Stirring was manual, so shear rate is not a controlled quantity.

Stage 3 · Casting and cure

Eleven days from casting to finished specimen, without thermal input



Mould material

Silicone, flexible enough to release without a parting agent and without damaging a partially consolidated specimen. 50 mm cube moulds for compression, flat moulds otherwise.

Thermal input

None after the 55-minute preparation. The entire cure proceeds at room temperature, which is relevant to the embodied-energy argument.

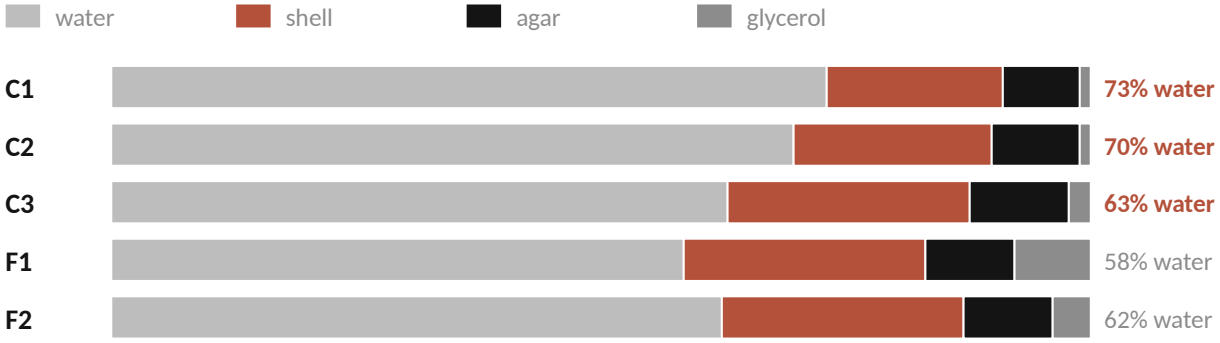
Limitation the composite shrinks appreciably as water evaporates. Cross-sections were measured at test time, but post-drying dimensions were not logged, so every dimension quoted in this presentation is a mould dimension.

Formulations

Five mixes, with the derived ratios that govern the results

ID	Test	Water g	Shell g	Agar g	Glycerol g	Solids g	Shell : binder	Solids : water	Glycerol % solids
C1	Compression	325	80	35	5	120	**2.00**	0.37	4.2
C2	Compression	310	90	40	5	135	**2.00**	0.44	3.7
C3	Compression	280	110	45	10	165	**2.00**	0.59	6.1
F1	Flexure	450	190	70	60	320	1.46	0.71	18.8
F2	Flexure	480	190	70	30	290	1.90	0.60	10.3

Batch composition by mass



Three specimens per formulation. Values reported later are means; individual values were not retained, so standard deviations cannot be given.

Compressive series

Shell-to-binder is held at exactly 2.00 across all three mixes, so solids concentration is the only variable — the water band shortens from C1 to C3.

Flexural series

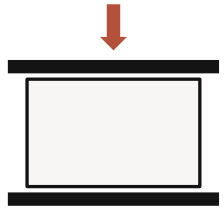
Glycerol was halved at fixed batch mass, so water rose with it. The plasticiser effect is not isolated.

Test programme

Five properties, three published standards, four specimen geometries

Property	Standard	n	Quantity measured
Compressive behaviour	Cube compression	3 per mix	Maximum load; compressive stress
Flexural behaviour	ASTM D790-17	3 per mix	Flexural strength; flexural modulus
Reaction to fire	ISO 11925-2:2020	2	Time to ignition; time to extinguish
Water absorption	ASTM D570-98(2018)	1	Mass gain over 24 h immersion
Bulk density	Gravimetric	1	Mass per mould volume

Mechanical values are means of three specimens; individual values were not retained, so standard deviations cannot be reported. Ignitability, absorption and density rest on single determinations and are indicative rather than characteristic values.



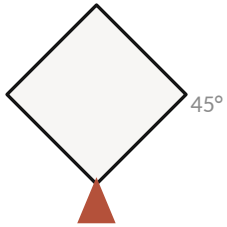
Cube compression

50 mm cube, platen-loaded



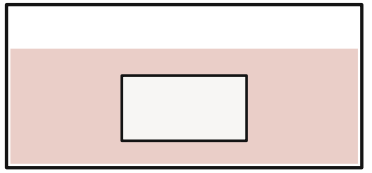
Three-point flexure

180 × 60 × 20 mm, 120 mm span



Edge ignition

40 × 50 mm at 45°, 15 s flame



Immersion

40 × 50 mm, shellac-coated, 24 h

All testing: Advanced Construction Materials Research Laboratory, Hacettepe University. · Schematic, not to scale.

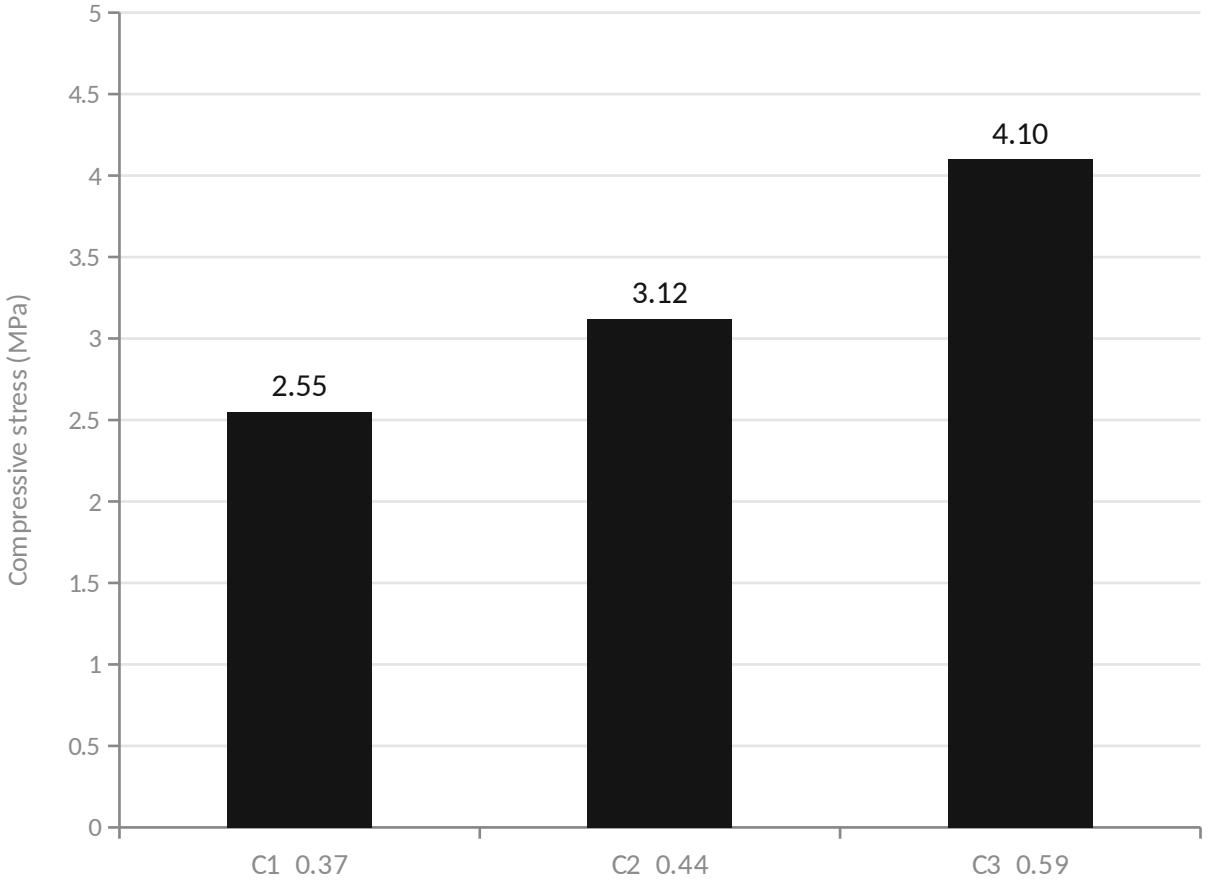
Compression

Measured values

ID	Solids : water	F_{max} kN	σc MPa
C1	0.37	3.89	2.55
C2	0.44	4.30	3.12
C3	0.59	**6.80**	**4.10**

Compressive strength rises with solids concentration, not with mineral content.

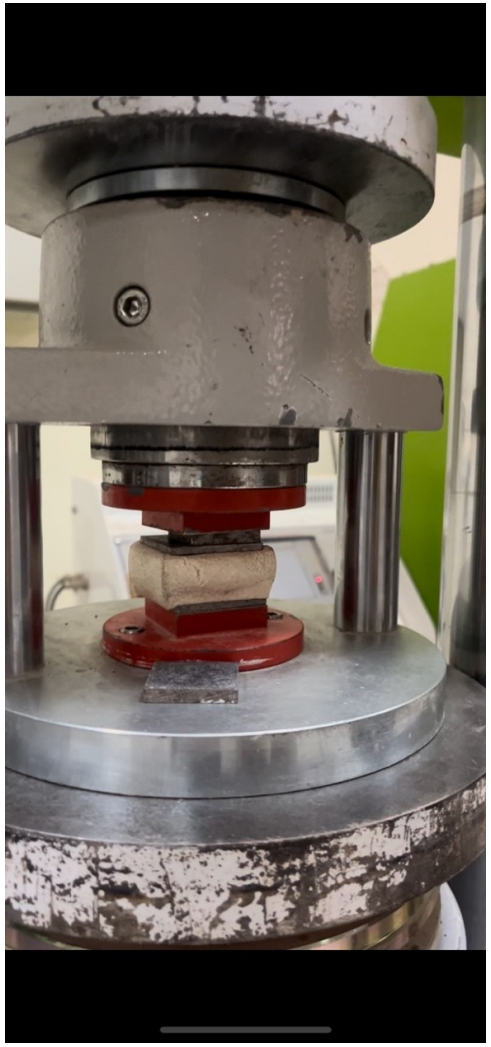
Shell-to-binder is fixed at 2.00 across the series, so packing density is the only variable. The relation is linear over the range tested, R^2 = 0.99 — three formulations, and therefore a hypothesis rather than a law.



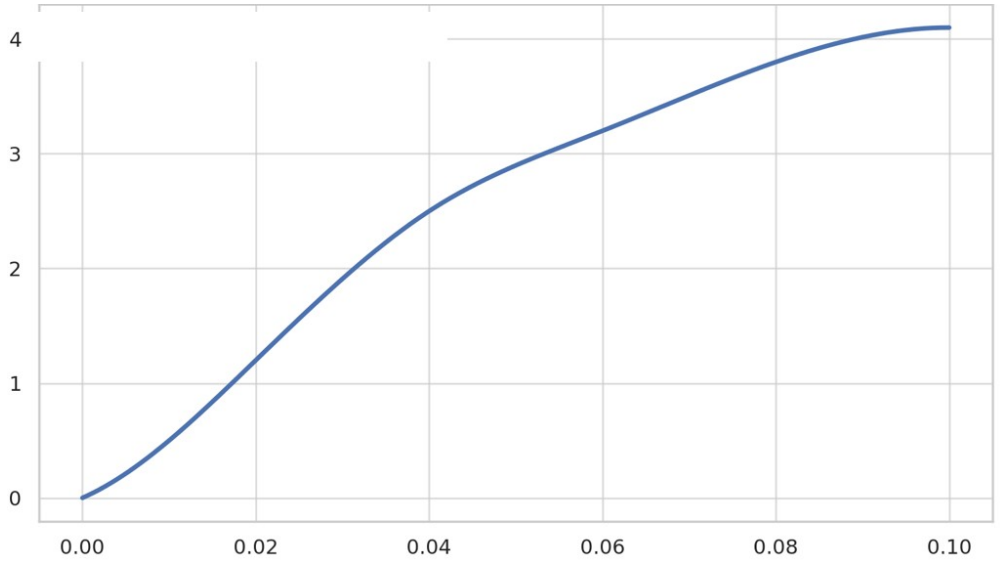
Fitted relation: σc = 6.93 (solids : water) + 0.04

Solids : water ratio shown beneath each formulation identifier.

Compressive failure



Compressive stress (MPa) – strain, C3



Observed response

- Linear to approximately 1 MPa, then progressive loss of stiffness
- Peak stress near 10% strain, with no abrupt load drop
- Continued deformation beyond peak with retained integrity

Interpretation

The agar network yields and the mineral redistributes rather than the section parting. Energy is absorbed rather than released, and the specimen remains a single body after test.

Failure proceeds by progressive crushing rather than by fracture.

For a demountable component this is the behaviour that governs service life: damage accumulates visibly rather than arriving as sudden loss of the section.

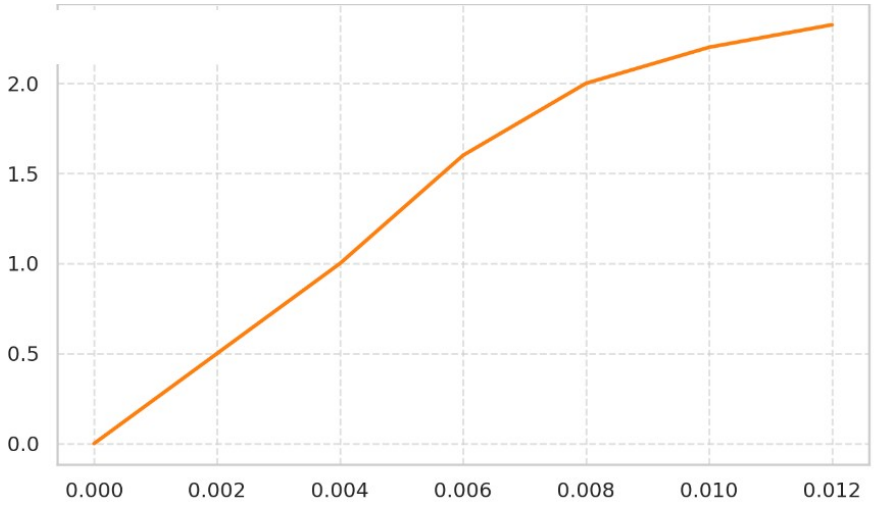
Strain derived from an assumed 10% maximum deformation based on observation, not instrumented measurement. · Bulk density 1.38 g/cm^3, mould-volume basis.

Flexure

Measured values — ASTM D790-17, three-point

ID	Glycerol % solids	F_{max} N	σF MPa	EF MPa	ε peak
F1	18.8	—	*not measurable*	—	—
F2	10.3	310	2.325	192	0.0121

Flexural stress (MPa) – strain, F2



Flexural strength computed as $\sigma_F = 3FL / 2bd^2$ on the full rectangular section.

At 18.8% glycerol the instrument registered no usable resistance. Halving the plasticiser gave a section that resists while still deforming substantially.

The transition lies between these two values.

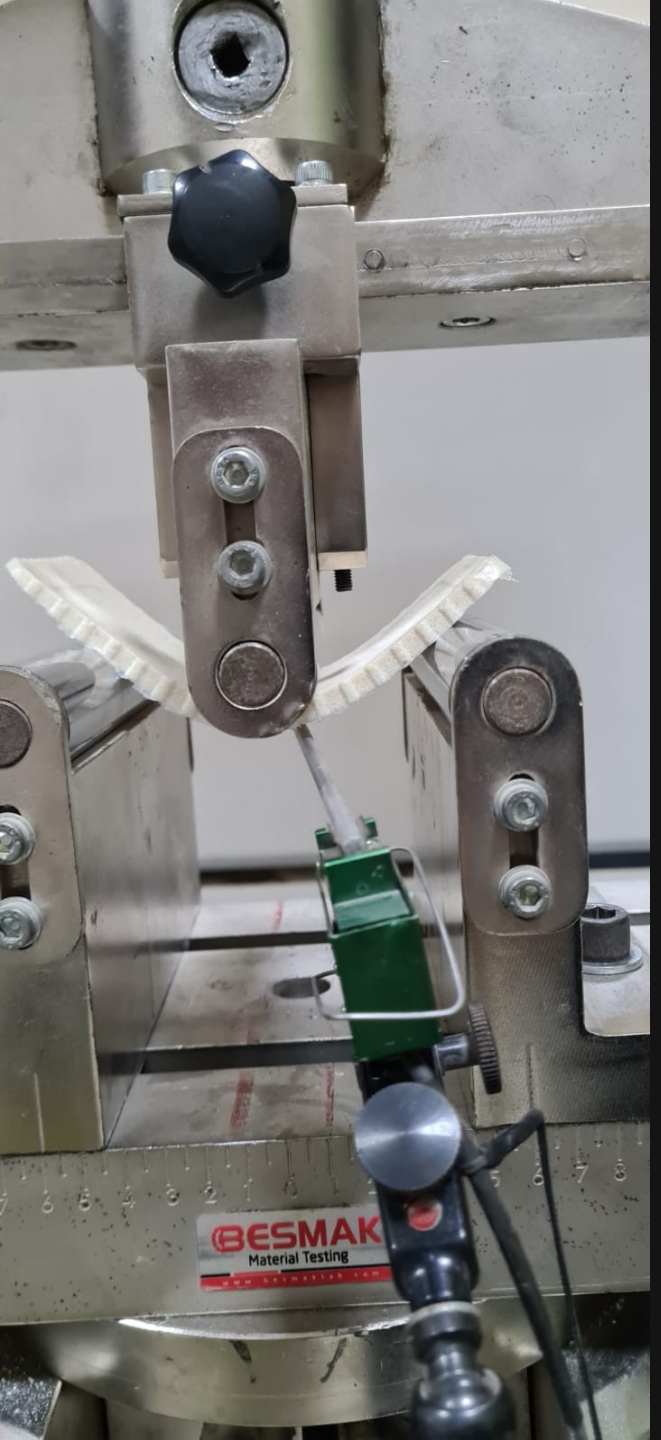
Limitation

The specimen is kerfed on its tension face. The standard assumes a solid rectangular section and the calculation uses the full section, so part of the measured compliance is geometric and the reported modulus is a lower bound.

Flexural resistance is governed by plasticiser content.

The composition at which the composite ceases to resist and merely deforms lies between 10.3 and 18.8% glycerol on solids. Locating it is the first task of the next formulation series.

Confounded with solids concentration: batch mass was held constant, so water rose as glycerol fell.







Flexural deformation

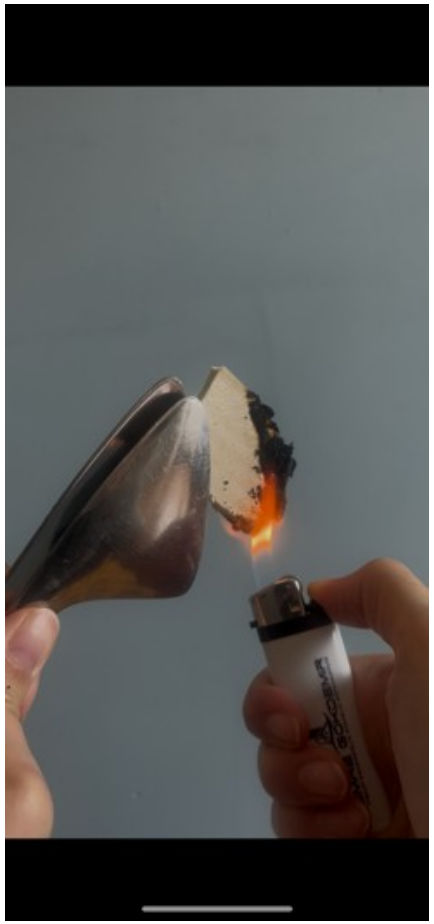
The beam is deflected well beyond the failure point of a comparable ceramic without parting. Transverse kerfs on the tension face permit the panel to be arched by hand.

The kerfs are also why the measured compliance is partly geometric, and why the reported modulus is a lower bound rather than a material constant.

Substantial deformation with retained integrity, in both test programmes.

Three-point rig at high deflection · close-up of the kerfed tension face · panel arched by hand

Fire, water, density



Ignitability

≈ 10 s

to ignition under direct flame

4 s

to self-extinguish after flame removal

- Charring remains local
- No lateral flame spread
- No melting and no dripping

The probable mechanism is endothermic decomposition of the carbonate fraction, diluting the flame front. This was not verified calorimetrically.

ISO 11925-2:2020. Ignitability only: no Euroclass classification, cone calorimetry, smoke or toxicity data.

Water absorption, 24 h immersion

0.7%

mass gain, shellac-coated. Coating integrity rather than the composite is the limiting factor for humid service.

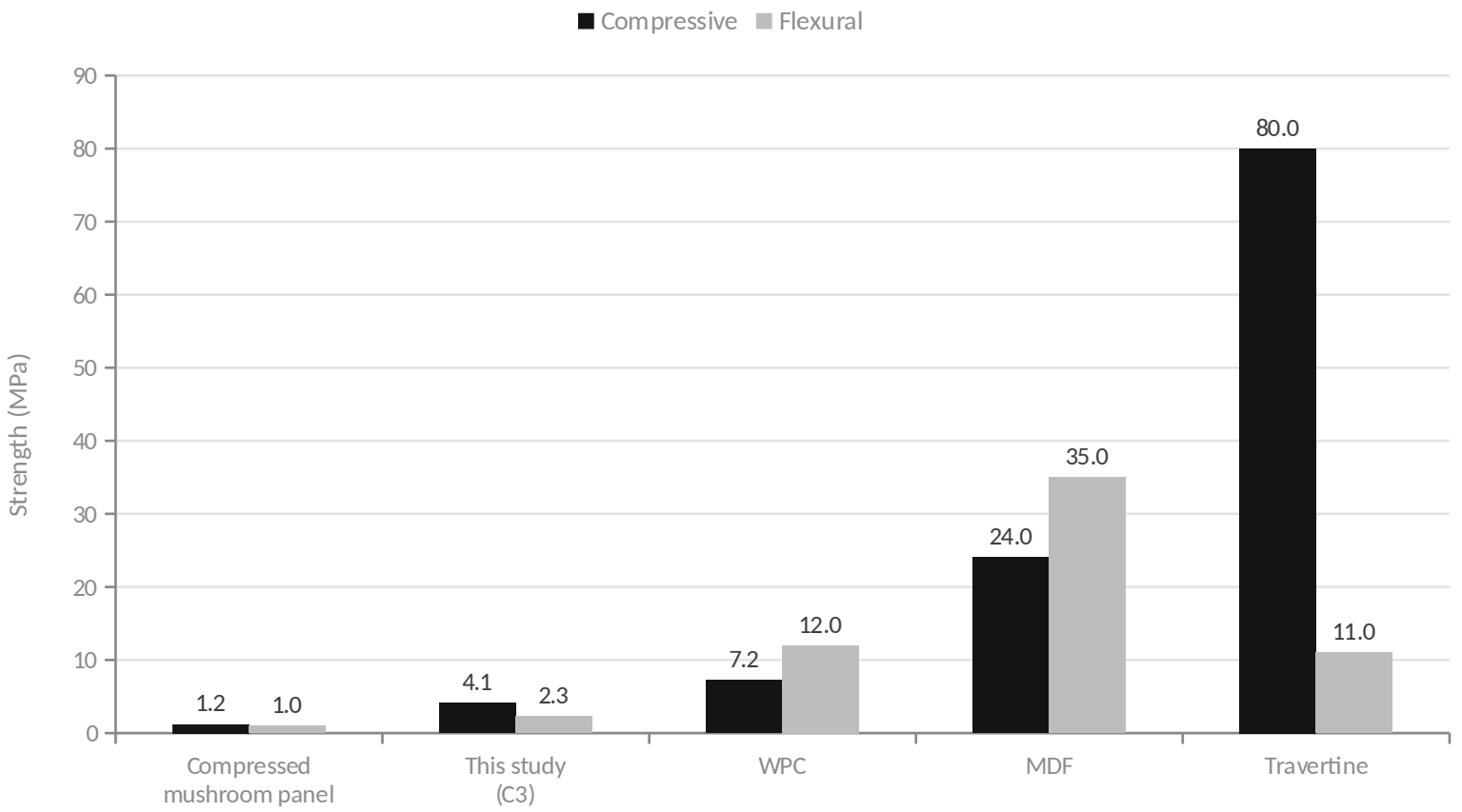
Bulk density

Material	g/cm³	Reference
This study (C3)	1.38	—
Mycelium, cellulose	0.3 – 0.8	Jones et al. (2020)
Fired clay brick	1.6 – 2.0	William & Owen (1995)
Normal-weight concrete	2.4	William & Owen (1995)

Denser than the fungal and cellulosic family, lighter than fired clay. Mould-volume basis, therefore a lower bound.

Conventional materials

Compressive and flexural strength against interior surface materials



Velasco & Gaona (2011) · Klyosov (2007) · Nemli & Aydın (2007) · Altun (2006)

Position

One sixth the compressive strength of MDF and one twentieth that of travertine, a stone the material visually resembles. Three times the compressive strength of compressed mushroom panel, the only other fully bio-based board in the set.

On strength alone the composite is not competitive with engineered wood or stone.

Bio-based composites

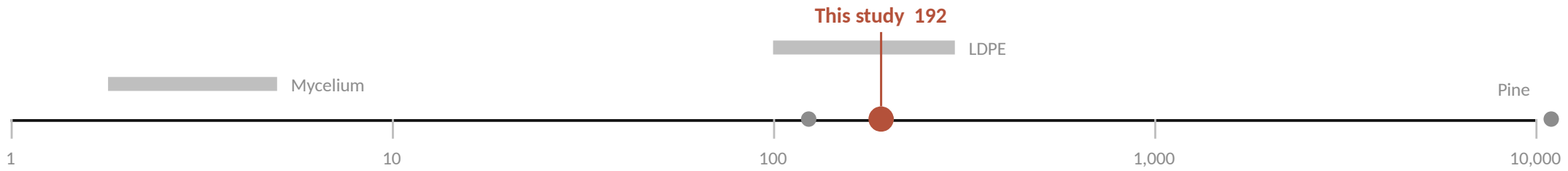
Compressive strength

Material	σc MPa	Reference
This study (C3)	**4.1**	—
Mycelium composite	0.145 – 0.452	Jones et al. (2020)
Hempcrete	0.3 – 0.6	Arnaud & Gourlay (2012)
Eggshell-filled PLA	2 – 5	Kim & Pal (2010)
Mineral-filled biocomposite	2.1 – 39.6	Shah et al. (2016)

Elastic modulus

Material	E MPa	Reference
This study (F2)	**192**	—
Mycelium composite	1.8 – 5	Islam et al. (2018)
Soy-based bioplastic	124	Mansour et al. (2021)
Flax-fibre PLA	198	Faruk et al. (2012)
LDPE	100 – 300	Zhao et al. (2022)
Wood (pine)	11 000	Pope & Hudson (1989)

Elastic modulus on a logarithmic scale, MPa

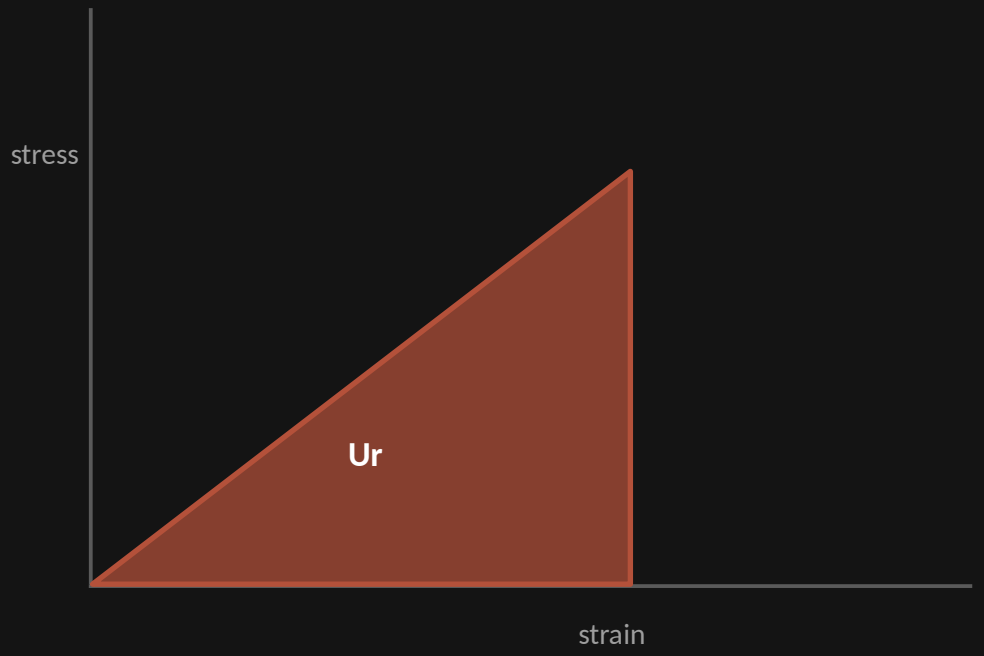


Water-cast and ambient-dried, the composite attains the stiffness range of melt-processed bio-polymers.

Unlabelled points: soy-based bioplastic 124 MPa, flax-fibre PLA 198 MPa. · An order of magnitude above the fungal and lime-binder family in compression; wood remains two orders of magnitude above in modulus.
Values are reported as published; specimen geometry and test method differ between sources.

Modulus of resilience

Elastic energy absorbed per unit volume



The shaded area is the energy absorbed per unit volume before permanent damage: ½ σ ε, and with ε = σ / E it reduces to the expression at right.

$$Ur = \sigma^2 / 2E$$

$$14.1\ kJ/m^3$$

computed from this study's own σ_F = 2.325 MPa and E_F = 192 MPa

Elastic modulus appears in the denominator.

The low modulus that counted against the material on the preceding slide is the same property that raises this quantity. Strength records the load at which performance ceases; it does not record whether failure is graceful or catastrophic.

Limitation

A comparative table cannot yet be assembled: the literature reports strength and modulus in separate studies on separate formulations.

This is a data point and a proposed reporting convention, not a ranking, and the value inherits the lower-bound modulus.

Resilience as a property of systems that absorb disturbance and retain function, after Holling (1973).

Application envelope

Modular proposal derived from the measurements



Demountable modules on a reversible fixing, replaceable one unit at a time.

No life-cycle assessment was carried out. These are qualitative potentials grounded in the protocol, not quantified claims.

Conditions of use, each traced to a measured result

Non-loadbearing surfaces

self-weight and handling loads only; σc = 4.1 MPa

Dry interiors

not wet rooms, not façades; absorption depends on coating integrity

Away from ignition sources

self-extinguishing, but unclassified for reaction to fire

Modular rather than sheet

damage is replaced one unit at a time

End-of-life pathway

Every constituent, the shellac finish included, is biodegradable; reversible fixing allows recovery of the module intact. What is displaced is quarried limestone, in a surface application rather than a downcycled one.

Limitations

Three classes, each with a defined remedy

Measurement

- No dispersion data — means of three specimens, individual values not retained
- Compressive strain assumed from observation, not instrumented
- Shrinkage unquantified, so bulk density is a lower bound

Process

- Flexural series confounded — glycerol and solids concentration moved together
- Kerfed section calculated as solid, so flexural modulus is a lower bound
- Manual stirring; a single laboratory and a single operator

Untested

- Durability — humidity, thermal cycling, UV, microbial attack, ageing
- Reaction-to-fire classification beyond ignitability
- Life-cycle assessment

Each limitation above corresponds to a defined next experiment, set out on the following slide.

Further work

Four experiments, in order of what each one resolves

01	**Repeat with dispersion reported**	Three or more specimens per formulation with individual values retained, so that standard deviations and a coefficient of variation can be reported.
02	**Record post-drying geometry**	Measure and log the dried cross-section, converting every lower bound in this presentation into a true stress and a true density.
03	**Instrument the strain**	Replace the assumed 10% deformation with measured strain, yielding a real stress–strain curve and a defensible modulus of resilience.
04	**Test an un-kerfed section**	Separate the geometric contribution from the material contribution, so that the flexural modulus becomes a material constant.

Beyond these four: humidity and thermal cycling, microbial resistance, a reaction-to-fire classification, and a life-cycle assessment to support the environmental argument quantitatively.

Conclusions

01	**A protocol that can be repeated**	every production parameter reported
02	**Packing density governs compressive strength**	not mineral content
03	**Plasticiser content sets brittle or ductile response**	the transition is sharp
04	**Bio-polymer stiffness without thermal processing**	water-cast, ambient-dried
05	**Strength alone is an insufficient criterion**	report σ and E together

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Testing: Advanced Construction Materials Research Laboratory, Hacettepe University, with thanks to Dr Hüseyin İlcan. Derived from the author's doctoral dissertation, Hacettepe University, 2025.

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